Learning Objective 9.5: I can determine the reflection and transmission of UPW at a boundary for Transverse Electric (TE) and Transverse Magnetic (TM) waves, using boundary conditions.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem the boundary is the plane $z = 0$, medium 1 is $z < 0$, medium 2 is $z > 0$, the incident wave travels in $+z$ with $\widetilde{\mathbf{E}}_i = \hat{\mathbf{x}} E_0 e^{-j\beta_1 z}$, and every medium is nonmagnetic, $\mu = \mu_0$.

1. **Deriving the coefficients.** Two lossless media meet at $z = 0$, with intrinsic impedances η_1 and η_2.

   (a) Write the electric and magnetic fields of the incident, reflected, and transmitted waves in terms of E_0, Γ, and τ.

$$\widetilde{\mathbf{E}}_i = \hat{\mathbf{x}} E_0 e^{-j\beta_1 z}, \qquad \widetilde{\mathbf{H}}_i = \hat{\mathbf{y}} \frac{E_0}{\eta_1} e^{-j\beta_1 z}$$

$$\widetilde{\mathbf{E}}_r = \hat{\mathbf{x}} \Gamma E_0 e^{+j\beta_1 z}, \qquad \widetilde{\mathbf{H}}_r = -\hat{\mathbf{y}} \frac{\Gamma E_0}{\eta_1} e^{+j\beta_1 z}$$

$$\widetilde{\mathbf{E}}_t = \hat{\mathbf{x}} \tau E_0 e^{-j\beta_2 z}, \qquad \widetilde{\mathbf{H}}_t = \hat{\mathbf{y}} \frac{\tau E_0}{\eta_2} e^{-j\beta_2 z}$$

   Each magnetic field is $\frac{1}{\eta}\hat{\mathbf{k}} \times \widetilde{\mathbf{E}}$. The reflected wave has $\hat{\mathbf{k}} = -\hat{\mathbf{z}}$, and $-\hat{\mathbf{z}} \times \hat{\mathbf{x}} = -\hat{\mathbf{y}}$.

   (b) Apply the continuity of tangential **E** and tangential **H** at $z = 0$, and solve for Γ and τ.

$$E_0 + \Gamma E_0 = \tau E_0 \implies \tau = 1 + \Gamma$$

$$\frac{E_0}{\eta_1}(1 - \Gamma) = \frac{\tau E_0}{\eta_2} \implies \eta_2 (1 - \Gamma) = \eta_1 (1 + \Gamma)$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \qquad \tau = \frac{2\eta_2}{\eta_2 + \eta_1}$$

   (c) Show that the reflected and transmitted fractions of the incident power, $|\Gamma|^2$ and $(\eta_1/\eta_2)\,|\tau|^2$, add to one.

$$|\Gamma|^2 + \frac{\eta_1}{\eta_2}|\tau|^2 = \frac{(\eta_2 - \eta_1)^2 + 4\eta_1\eta_2}{(\eta_2 + \eta_1)^2} = \frac{\eta_2^2 + 2\eta_1\eta_2 + \eta_1^2}{(\eta_2 + \eta_1)^2} = 1$$

2. **Air to glass.** A 3 GHz plane wave with $E_0 = 10$ V/m travels in air and strikes glass with $\epsilon_r = 4$.

(a) Find η_2, Γ, and τ.

$$\eta_2 = \frac{376.7}{\sqrt{4}} = 188.4\,\Omega$$
$$\Gamma = \frac{188.4 - 376.7}{188.4 + 376.7} = -\frac{1}{3}, \qquad \tau = \frac{2}{3}$$

Γ = -0.333

τ = 0.667

(b) Find the amplitudes of the incident, reflected, and transmitted magnetic fields, and show that tangential **H** is continuous.

$$\frac{E_0}{\eta_1} = \frac{10}{376.7} = 26.5\text{ mA/m}$$
$$-\frac{\Gamma E_0}{\eta_1} = +8.85\text{ mA/m}$$
$$\frac{\tau E_0}{\eta_2} = \frac{6.67}{188.4} = 35.4\text{ mA/m}$$

All three point along $\hat{\mathbf{y}}$ at $z = 0$, and $26.5 + 8.85 = 35.4$ mA/m.

H_t = 35.4 mA/m

(c) Find the SWR in the air, the largest and smallest field magnitudes, and the distance from the boundary to the nearest maximum.

$$\text{SWR} = \frac{1 + 1/3}{1 - 1/3} = 2$$
$$E_{\text{max}} = 13.3\text{ V/m}, \qquad E_{\text{min}} = 6.67\text{ V/m}$$

$\Gamma < 0$ puts a minimum at the boundary. With $\lambda_1 = c/f = 10.0$ cm, the nearest maximum is $\lambda_1/4 = 2.50$ cm into the air.

SWR = 2.0

max at = $z = -2.50$ cm

(d) Find the incident power density and the percentages of the power that reflect and transmit.

$$S_i = \frac{E_0^2}{2\eta_1} = \frac{100}{753.5} = 0.133\text{ W/m}^2$$
$$|\Gamma|^2 = 11.1\%, \qquad \frac{\eta_1}{\eta_2}|\tau|^2 = 2\left(\frac{4}{9}\right) = 88.9\%$$

reflected = 11.1%

3. **Glass to air.** The same boundary, now with the wave arriving from inside the glass: medium 1 has $\epsilon_r = 4$, and medium 2 is air.

   (a) Find Γ and τ.

$$\Gamma = \frac{376.7 - 188.4}{376.7 + 188.4} = +\frac{1}{3}, \qquad \tau = \frac{4}{3}$$

τ = 1.33

   (b) The transmitted electric field is larger than the incident one. Explain why that does not create power, and find the fraction of the power transmitted.

   The transmitted wave travels in the medium with the larger impedance, so it carries less power per unit of $|E|^2$:

$$\frac{S_t}{S_i} = \frac{\eta_1}{\eta_2}|\tau|^2 = \frac{1}{2}\left(\frac{16}{9}\right) = 88.9\%$$

   The same 11.1% reflects in both directions. The magnetic field shows the difference: $\tau E_0/\eta_2$ is smaller than the incident E_0/η_1.

   (c) Where is the field in the glass largest, and why does the answer differ from Question 2?

   $\Gamma > 0$, so the incident and reflected fields add in phase at the boundary, and the maximum sits at $z = 0$. A higher impedance behind the boundary acts like a load larger than Z_0 in Lesson 6.

4. **A conducting wall.** A 10 GHz plane wave with $E_0 = 5$ V/m (real) travels in air and strikes a flat perfect conductor that fills $z > 0$.

   (a) Use the boundary condition on tangential **E** to find Γ, and write the total phasor fields in the air.

   E $= 0$ in the conductor, so $E_0(1+\Gamma) = 0$ and $\Gamma = -1$.

   $$\tilde{\mathbf{E}}_1 = \hat{\mathbf{x}} E_0 \left(e^{-j\beta_1 z} - e^{+j\beta_1 z}\right) = -\hat{\mathbf{x}}\, 2jE_0 \sin\beta_1 z$$
   $$\tilde{\mathbf{H}}_1 = \hat{\mathbf{y}} \frac{E_0}{\eta_1}\left(e^{-j\beta_1 z} + e^{+j\beta_1 z}\right) = \hat{\mathbf{y}} \frac{2E_0}{\eta_1} \cos\beta_1 z$$

   (b) Find the distance from the wall to the nearest maximum of $|\mathbf{E}|$, and its value.

   $\lambda_1 = c/f = 3.00$ cm. The maximum of $|\sin\beta_1 z|$ nearest the wall is at $\lambda_1/4 = 0.750$ cm, where $|\mathbf{E}| = 2E_0 = 10$ V/m.

   distance = 0.750 cm

   (c) Find the surface current density on the wall.

   With $\hat{\mathbf{n}} = -\hat{\mathbf{z}}$, from the conductor into the air, and $\mathbf{H}_2 = 0$:

   $$\mathbf{J}_s = \hat{\mathbf{n}} \times \tilde{\mathbf{H}}_1(0) = -\hat{\mathbf{z}} \times \hat{\mathbf{y}} \frac{2E_0}{\eta_1} = \hat{\mathbf{x}}\, 26.5 \text{ mA/m}$$

   $\mathbf{J}_s$ = $\hat{\mathbf{x}}\, 26.5$ mA/m

   (d) Write $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$, and explain why no power flows toward the wall on average.

   $$\mathbf{E}(z,t) = \text{Re}\left[\tilde{\mathbf{E}}_1 e^{j\omega t}\right] = \hat{\mathbf{x}}\, 2E_0 \sin(\beta_1 z)\sin(\omega t)$$
   $$\mathbf{H}(z,t) = \hat{\mathbf{y}}\, \frac{2E_0}{\eta_1} \cos(\beta_1 z)\cos(\omega t)$$

   The two fields are a quarter period apart in time, so the Poynting vector $\mathbf{E} \times \mathbf{H} \propto \sin\omega t \cos\omega t$ averages to zero. The reflected wave carries back all the power the incident wave brings.

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